
Department of Civil Engineering, IIT Madras
CV2090 (Probability & Statistics for Civil Engineers)
Jul–Nov 2026

28 Sep, 2026

Tutorial-6

3:00–5:00 pm

Instructions:

1. Make assumptions wherever necessary, and *state* them explicitly.
 2. Tutorials are closed-phone, closed-tablet, and closed-LLM.
 3. You can bring books and notes, you are welcome to discuss with friends and TAs. However, your submission should be your own work.
 4. Submit handwritten solutions on A4 sheets, prepared during the tutorial hours.
 5. Submissions will not be accepted after the tutorial hours.
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Parameter Estimation and Confidence Intervals

Q1. Method of Moments Estimation (MoM) The probability density function (PDF) of a Rayleigh distributed random variable X with scale parameter $\alpha > 0$ is given by:

$$f_X(x; \alpha) = \frac{x}{\alpha^2} \exp\left(-\frac{x^2}{2\alpha^2}\right), \quad x \geq 0$$

The theoretical mean and standard deviation of the Rayleigh distribution in terms of α are:

$$\mathbb{E}[X] = \alpha\sqrt{\frac{\pi}{2}}, \quad \sigma_X = \alpha\sqrt{2 - \frac{\pi}{2}}$$

A total of $n = 10$ measurements of maximum wind speeds (m/s) during storm events were recorded as follows:

12.1, 14.3, 10.8, 15.2, 11.5, 13.0, 12.7, 14.0, 11.8, 13.6

- (a) Calculate the sample mean $\bar{x}$ and sample standard deviation s for the dataset.
- (b) Estimate the scale parameter α using the MoM by equating $\bar{x}$ to $\mathbb{E}[X]$.
- (c) Using $\hat{\alpha}_{\text{MoM}}$, compute the theoretical standard deviation σ_X of the fitted Rayleigh model, compare it with s , and explain the discrepancy. ■

Q2. Properties of Point Estimators Let $\hat{\theta}$ be an estimator of an unknown parameter θ derived from an i.i.d. sample $X_1, X_2, \dots, X_n$ with mean μ and variance σ^2 .

- (a) Define the bias $B(\hat{\theta})$ and state the condition for an estimator to be unbiased.
- (b) Show that the sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is an unbiased estimator of μ .
- (c) Consider two competing estimators for the population variance σ^2 :

$$\hat{\sigma}_1^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \quad \text{and} \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Calculate the expected value for both estimators, determine their biases, and explain the role of the correction factor $\left(\frac{n}{n-1}\right)$. ■

Q3. Maximum Likelihood Estimation (MLE) for Continuous Distributions Let $X_1, X_2, \dots, X_n$ be an i.i.d. random sample with likelihood function $\mathcal{L}(\boldsymbol{\theta}) = \prod_{i=1}^n f_X(x_i; \boldsymbol{\theta})$.

- (a) **Exponential Distribution:** Given $f_X(x; \lambda) = \lambda e^{-\lambda x}$ ($x \geq 0, \lambda > 0$), derive the MLE $\hat{\lambda}_{\text{MLE}}$ in terms of $\bar{X}$.
- (b) **Normal Distribution:** Given $X_i \sim N(\mu, \sigma^2)$, derive the simultaneous MLEs $\hat{\mu}_{\text{MLE}}$ and $\hat{\sigma}_{\text{MLE}}^2$.

Hint: Maximize the log-likelihood function $\ln \mathcal{L}(\boldsymbol{\theta}) = \sum_{i=1}^n \ln f_X(x_i; \boldsymbol{\theta})$ by setting $\frac{\partial \ln \mathcal{L}}{\partial \theta_j} = 0$. ■

Q4. Two-Sided and One-Sided Confidence Intervals for Population Mean A batch of $n = 10$ steel specimens underwent Charpy V-notch impact testing. The measured impact energy values (Joules) are:

64.1, 64.7, 64.5, 64.6, 64.5, 64.3, 64.6, 64.8, 64.2, 64.3

The sample mean is $\bar{x} = 64.46$ J. Assume the population is normally distributed with known standard deviation $\sigma = 1.0$ J.

- (a) Construct a **95% two-sided confidence interval** for the population mean μ .
- (b) Construct a **95% lower confidence bound** for μ .
- (c) Construct a **95% upper confidence bound** for μ .
- (d) Provide two practical engineering examples for each of the three types of confidence intervals/bounds: two-sided confidence intervals, lower confidence bounds, and upper confidence bounds. ■

Q5. Confidence Intervals for Variance, Standard Deviation, and Small-Sample Mean Refer to the dataset from Q4 ($\bar{x} = 64.46$ J, $n = 10$). Now assume that the population standard deviation σ is **unknown**.

- (a) Calculate the sample variance s^2 and sample standard deviation s .
- (b) Construct a **95% two-sided confidence interval for σ^2** using the Chi-Square (χ^2) distribution.
- (c) Deduce the corresponding **95% two-sided confidence interval for σ** .
- (d) Construct a **95% two-sided confidence interval for μ** using Student's t -distribution. Compare the width of confidence interval from Q4(a) and explain the reason for any difference.

Note: For $v = 9$ degrees of freedom, the Chi-Square critical values are $\chi_{0.975,9}^2 = 2.700$ and $\chi_{0.025,9}^2 = 19.023$, and the Student's t critical values are $t_{0.025,9} = 2.262$ and $t_{0.05,9} = 1.833$. ■